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4 Recursive Competitive Equilibrium

So far we looked at individuals' decision problems. Now we want to look at an economy with the representative consumer and the representative firm in a competitive equilibrium. We want to use recursive methods to define a competitive equilibrium in this economy.

Growth Model with Leisure

The consumer faces the following problem

$$\begin{aligned} \max_{\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t) \\ \text{s.t.} \quad & c_t + k_{t+1} - (1 - \delta)k_t = w_t^* n_t + r_t^* k_t, \quad \forall t \\ & k_{t+1} \geq 0, \quad c_t \geq 0, \quad 0 \leq n_t \leq 1, \quad \forall t \\ & k_0 \text{ given.} \end{aligned}$$

Assume that the representative firm has a CRS production technology $F(k, n)$ so that the profits are zero.

This problem *appears* to land nicely into a recursive formulation:

$$\begin{aligned} v(k) = \max_{n, k'} \quad & \{U(w^* n + r^* k + (1 - \delta)k - k', 1 - n) + \beta v(k')\} \\ \text{s.t.} \quad & 0 \leq k' \leq w^* n + r^* k + (1 - \delta)k \\ & 0 \leq n \leq 1. \end{aligned}$$

However, this is problematic because of the prices w^* and r^* , which are not constant over time.

Issue: How do we represent prices recursively?

Idea: Define a “state of the economy” or an “aggregate state” and represent prices as functions of this state.

Let S_t be the aggregate state at time t , and $w(S_t), r(S_t)$ be the pricing functions. Last, assume a “law of motion” for S_t : $S_{t+1} = \Lambda(S_t)$.

Now given functions $w(S_t), r(S_t), \Lambda(S_t)$ and so, we can generate $\{r_t^*\}, \{w_t^*\}$ by

$$w_0^* = w(S_0), \quad r_0^* = r(S_0)$$

$$S_1 = \Lambda(S_0), \quad w_1^* = w(S_1), \quad r_1^* = r(S_1)$$

$$S_2 = \Lambda(S_1), \dots$$

A “*Big K, little k*” trick: Introduce K as an aggregate state variable and k as an individual state variable.

Definition of a RCE

For the simple growth model, a RCE is a list of functions

- $v(k, K)$ - an individual value function of k , the individual capital, and K , the aggregate capital

- $k'(k, K)$ }
- $n(k, K)$ } - decision rules for individual capital tomorrow and labor today
- $K'(K)$ - capital tomorrow

- $k^f(K)$ }
- $n^f(K)$ } - capital demand and labor demand today

- $r(K)$ }
- $w(K)$ } - price functions (price of output is normalized to one),

such that

(1) (Consumer’s maximization)

Taking $r(K), w(K)$, and $K'(K)$ as given, $v(k, K)$ solves the following functional equation:

$$v(k, K) = \max_{n, k'} \{U(nw(K) + kr(K) + (1 - \delta)k - k', 1 - n) + \beta v(k', K'(K))\}$$

$$s.t. \quad 0 \leq k' \leq w(K)n + r(K)k + (1 - \delta)k$$

$$0 \leq n \leq 1$$

and $k'(k, K), n(k, K)$ are the optimal decision rules.

(2) (Firm Maximization)

Taking $r(K), w(K)$ as given, for each K , $k^f(K)$ and $n^f(K)$ solve

$$\begin{aligned} \max_{k,n} & F(k, n) - w(K)n - r(K)k \\ \text{s.t.} & \quad k \geq 0, n \geq 0 \end{aligned}$$

(3) (Consistency)

$$\begin{aligned} k &= K \\ K'(K) &= k'(K, K) \end{aligned}$$

(4) (Market Clearing)

$$\begin{aligned} K &= k^f(K) \\ n(K, K) &= n^f(K) \end{aligned}$$

Goods market clears by Walras' Law.

In this setup k is the individual state variable and K is the aggregate state variable. The role of the aggregate state variable is to represent current and future prices. Given an economy, if you want to define a RCE, you must first decide on individual state variables for all the individual agents, and the aggregate state variables. Given individual state variables, the general rule is that the aggregate state variable is the distribution of individual state variables across all individuals.

Production with Adjustment Costs

Consider the standard growth model with leisure, but consider a production function with adjustment costs: $f(k_t, k_{t-1}, n_t)$. The references for this model are Chapter 12 of Ljungqvist and Sargent (2012), Lucas and Prescott (1971), and Prescott and Mehra (1980).

Recall that the key feature of a state variable is that it summarizes something from the past.

State variables for this problem:

- Individual:

- HH: k, b
- Firm: k_{-1}
- Aggregate: K, B, K_{-1}

Recall that the aggregate state variables should be the distributions of all individual state variables.

A RCE for this economy is a list of functions

- | | | |
|-------------------------------|---------------------------------|----------------------|
| • $v^c(k, b, K, K_{-1}, B)$ | • $k^f(k_{-1}, K, K_{-1}, B)$ | • $B'(K, K_{-1}, B)$ |
| • $k'^c(k, b, K, K_{-1}, B)$ | • $n^f(k_{-1}, K, K_{-1}, B)$ | • $w(K, K_{-1}, B)$ |
| • $b'^c(k, b, K, K_{-1}, B)$ | • $\pi^f(k_{-1}, K, K_{-1}, B)$ | • $r(K, K_{-1}, B)$ |
| • $n^c(k, b, K, K_{-1}, B)$ | • $\Pi(K, K_{-1}, B)$ | • $q(K, K_{-1}, B)$ |
| • $v^f(k_{-1}, K, K_{-1}, B)$ | • $K'(K, K_{-1}, B)$ | • $q(K, K_{-1}, B)$ |

such that

(1) (Consumer maximization)

Taking $r(K, K_{-1}, B), w(K, K_{-1}, B), q(K, K_{-1}, B), K'(K, K_{-1}, B), B'(K, K_{-1}, B), \Pi(K, K_{-1}, B)$ as given, $v^c(k, b, K, K_{-1}, B)$ solves

$$v^c(k, b, K, K_{-1}, B) = \max_{n, k', b'} \{U(w(K, K_{-1}, B)n + r(K, K_{-1}, B)k + \Pi(K, K_{-1}, B) + (1 - \delta)k - k' + b - q(K, K_{-1}, B)b', 1 - n) + \beta v^c(k', b', K'(K, K_{-1}, B), K, B'(K, K_{-1}, B))\}$$

$$\begin{aligned} s.t. \quad & 0 \leq k' \leq w(K, K_{-1}, B)n + r(K, K_{-1}, B)k + \Pi(K, K_{-1}, B) + b + (1 - \delta)k - q(K, K_{-1}, B)b' \\ & 0 \leq n \leq 1 \\ & b' \geq \underline{b} \end{aligned}$$

and $n^c(k, b, K, K_{-1}, B), k'^c(k, b, K, K_{-1}, B), b'^c(k, b, K, K_{-1}, B)$ are the corresponding optimal decision rules.

(2) (Firm Maximization)

Taking $r(K, K_{-1}, B)$, $w(K, K_{-1}, B)$, $q(K, K_{-1}, B)$, $K'(K, K_{-1}, B)$, $B'(K, K_{-1}, B)$ as given, $v^f(k_{-1}, K, K_{-1}, B)$ solves

$$\begin{aligned} v^f(k_{-1}, K, K_{-1}, B) = \max_{k, n} \{ & f(k, k_{-1}, n) - w(K, K_{-1}, B)n - r(K, K_{-1}, B)k \\ & + q(K, K_{-1}, B)v^f(k, K'(K, K_{-1}, B), K, B'(K, K_{-1}, B)) \} \\ \text{s.t.} \quad & k \geq 0, \quad n \geq 0, \end{aligned}$$

and $k^f(k_{-1}, K, K_{-1}, B)$, $n^f(k_{-1}, K, K_{-1}, B)$ are the corresponding optimal decision rules, and $\pi(k_{-1}, K, K_{-1}, B) = f(k^f(\cdot), k_{-1}, n^f(\cdot)) - w(\cdot)n^f(\cdot) - r(\cdot)k^f(\cdot)$.¹

(3) (Consistency)

$$\begin{aligned} k &= K \\ k_{-1} &= K_{-1} \\ k'^c(K, 0, K, K_{-1}, 0) &= K'(K, K_{-1}, 0) \\ b'^c(K, 0, K, K_{-1}, 0) &= B'(K, K_{-1}, 0) \\ \pi(K_{-1}, K, K_{-1}, 0) &= \Pi(K, K_{-1}, 0) \end{aligned}$$

(4) (Market Clearing)

$$\begin{aligned} n^c(K, 0, K, K_{-1}, 0) &= n^f(K_{-1}, K, K_{-1}, 0) \\ K &= k^f(K_{-1}, K, K_{-1}, 0) \\ B'(K, K_{-1}, 0) &= 0 \end{aligned}$$

Goods market clears by Walras' Law.

References

LJUNGQVIST, L. AND T. J. SARGENT (2012): *Recursive Macroeconomic Theory*, MIT Press.

¹To understand why the firm's discount factor β^f must equal to the bond price, think about how much a dollar of the firm's earnings today is worth tomorrow. Saving $\hat{b}$ dollars today changes the firm's value by $-q(K, K_{-1}, B)\hat{b} + \beta^f\hat{b}$. Unless $\beta^f = q(K, K_{-1}, B)$, the firm could reach an infinite value by borrowing or saving.

LUCAS, R. E. AND E. C. PRESCOTT (1971): “Investment Under Uncertainty,” *Econometrica*, 39, 659–681.

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